# Ethan Wu

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## Education

**Queen's University** Kingston, ON

*Bachelor of Applied Science, Mechatronics and Robotics Engineering* *Sept. 2024 – April 2028*

– GPA: 3.89/4.3 | Dean's Scholar

## Experience

**Task Subteam Lead – NASA HERC RC Challenge** Sept. 2026 – Present

*Queen's Lunar Exploration Team (QLET)* *Kingston, ON*

– Leading the multidisciplinary task subteam for the 2027 NASA HERC competition, owning end-to-end design and execution of the autonomous task unit mechanism.
– Architecting the task execution state machine in ROS2 and C++ to parse incoming radio command sequences and trigger automated rover routines.
– Integrating a pan-tilt camera mechanism with OpenCV target recognition algorithms to isolate geometric shapes, calculate target distances via focal geometry, and dynamically apply digital color filters.
– Collaborating across mechanical and electrical subteams to prototype pan-tilt actuators, integrate optical sensors, and validate system power distribution.

**Embedded Systems Member** Sept. 2025 – June 2026

*Queen's AutoDrive Team* *Kingston, ON*

– Engineered a C++ memory monitoring module for an embedded watchdog to detect real-time system crashes.
– Earned 2nd Place in the Mobility Innovation Challenge at the SAE AutoDrive Competition for an AV adoption framework.
– Earned 3rd place overall and 3rd place for localization challenge for AutoDrive Y5.

**Makerspace Volunteer Supervisor** Sept. 2026 – Present

*SparQ Makerspace, Queen's University* *Kingston, ON*

– Supervise prototyping, providing technical instruction on FDM 3D printing, laser cutting, and circuit soldering.
– Maintain benchtop diagnostic equipment, power tools, and lab machinery while enforcing shop safety standards.

**Automation Team Manager** Oct. 2025 – Present

*Queen's Engineering Society* *Kingston, ON*

– Lead a development team constructing automated Power Automate flows and web interfaces to streamline organization data pipelines.
– Manage technical developer recruitment, project onboarding, and system architecture for administrative tools.

## Projects

**TriPAD - Construction Site Sensor** | *C++, ESP32, Sensor Fusion* Apr. 2026 – Aug. 2026

– Built a standalone environmental monitoring unit using an ESP32 to process particulate (PMS5003), noise (MAX9814), and IMU vibration (MPU6050) sensor telemetry over I2C and UART.
– Designed a 3D-printed Autodesk Fusion enclosure and won 3rd place at the General's Lab Innovation competition.

**Mars Sample Return (MSR) Rover** | *ROS2, C++, Python, R&D* Jan. 2026 – Apr. 2026

– Developed a ROS2-based autonomous rover with a bidirectional Raspberry Pi-to-Arduino communication bridge.
– Implemented closed-loop PI control with IMU/encoder sensor fusion and integrated RPLIDAR A1 for Nav2 localization.

**Prototype Gear-Based Launcher Turret (In-Progress)** | *C++, Arduino, Autodesk Fusion, Motor Drivers* Oct. 2026

– Designed and built a high-velocity pan-tilt BB launching turret utilizing DC motors, SG90/MG996R servos, DFRobot IO Expansion Shield v5, and DF-MD v1.3.
– Programmed Arduino Uno in C++ for closed-loop motor positioning and sensor-feedback control routines.
– Engineered custom structural mounting using everyday materials and 3D-printed parts in Autodesk Fusion.

## Technical Skills

**Software & Controls**: C/C++, Python, Java, Reinforcement Learning (RL), ROS2, Nav2, SLAM, OpenCV, PID/PI Control, Sensor Fusion, Kinematics, SQL, Linux, HTML, CSS
**Hardware & Embedded**: ESP32, Raspberry Pi, Arduino, VHDL, CAN bus, UART, I2C, PMS5003, MAX9814, MPU6050, LiDAR, Encoders
**CAD & Tools**: Autodesk Fusion, SOLIDWORKS, VHDL, LTSpice, Quartus, MATLAB, MS Power Automate, Git, LaTeX
**Hands-On**: Soldering, Circuit Debugging, Machining, Welding, Woodworking, 3D Printing; Languages: Cantonese, Mandarin, French